

CS4277 / CS5477

3D Computer Vision

Lecture 1: 2D and 1D Projective Geometry

Asst. Prof. Lee Gim Hee

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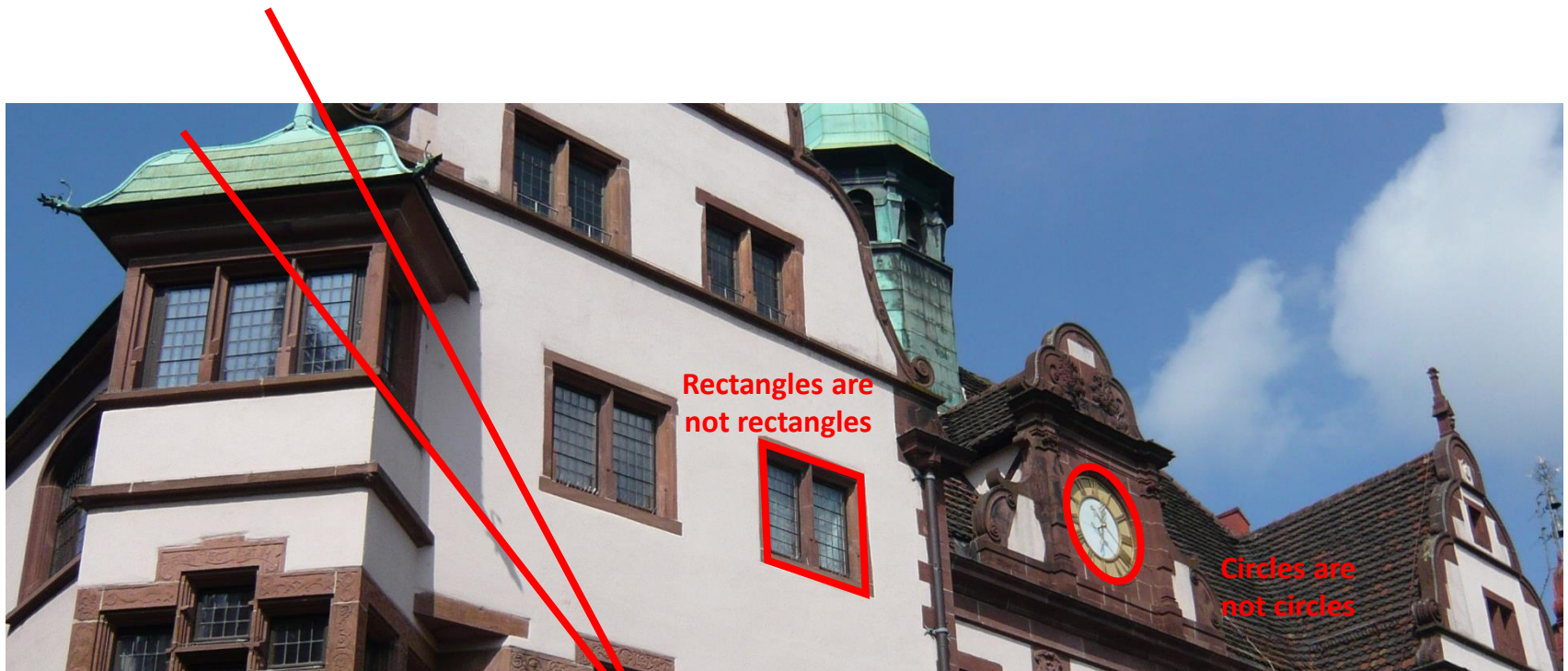
Semester 2

Acknowledgements

- A lot of slides and content of this lecture are adopted from:
 1. R. Hartley, and Andrew Zisserman: “Multiple view geometry in computer vision”, Chapter 2.
 2. Y. Ma, S. Soatto, J. Kosecka, S. S. Sastry, “ An invitation to 3-D vision”, Chapter 2.

Projective Transformation

- The **mapping of scene objects onto an image** is an example of a projective transformation.



Parallel lines meet
at a finite point

Rectangles are
not rectangles

Circles are
not circles

G. H. Lee "A random building", Freiburg, Germany, 2013.

What is Projective Geometry?

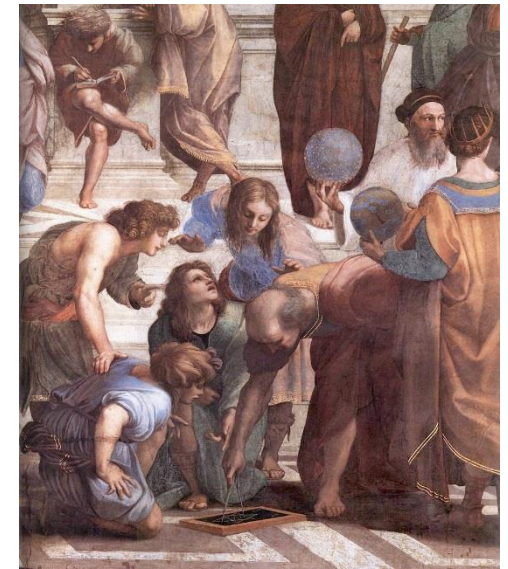
- We saw that certain geometric properties **are not preserved** by projective transformation, e.g.
 1. A circle may appear as an ellipse
 2. Parallel lines may meet at a finite point
 3. A rectangle may appear as a parallelogram
- In fact, **angles, distance, ratios of distances** – none of these are preserved!

What is Projective Geometry?

- A property that is preserved is **straightness**, which is the most general requirement on the mapping.
- **A thought:** we may define a projective transformation as any mapping that **preserves** straight lines.
- More generally, we study of geometric properties that are **invariant** with respect to projective transformations in projective geometry!

Euclidean vs Projective

- The familiar **Euclidean geometry** is an example of **synthetic geometry**.
- Use **axiomatic method** and its related tools, i.e. **compass and straightedge** to solve problems.
- **Projective geometry** uses coordinates and algebra – **analytic geometry**.
- We will see that one most important result is that **geometry at infinity** can now be nicely represented!



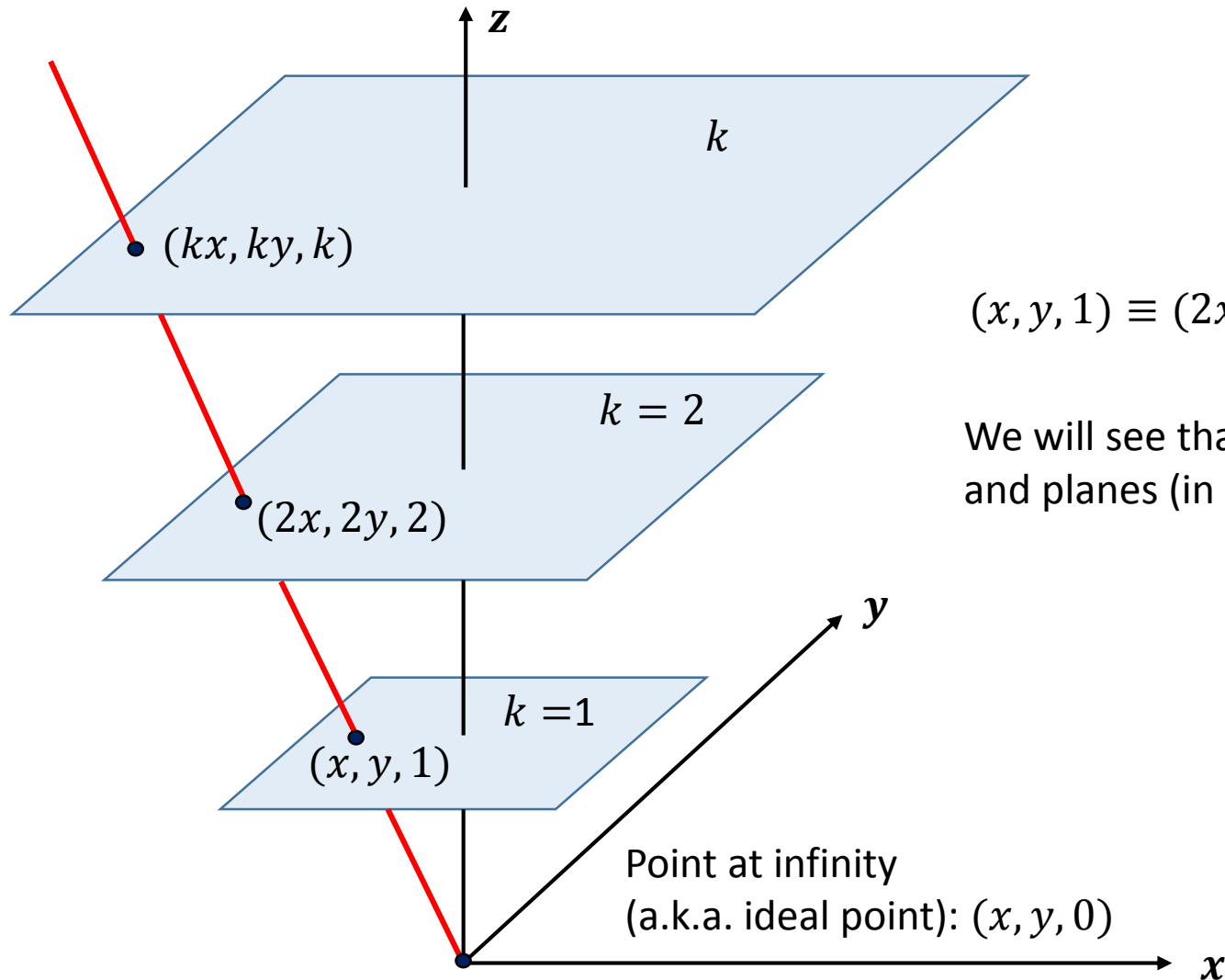
Raphael, "The School of Athens", 1509-1511

Image source: https://en.wikipedia.org/wiki/Euclidean_geometry

Homogenous Coordinates

- A point in homogenous coordinates (kx, ky, k) , corresponds to $\left(\frac{kx}{k}, \frac{ky}{k}\right) = (x, y)$ in **Cartesian coordinates**.
- (kx, ky, k) **is equivalent** for all k 's.
- Now we can use (x, y, k) , where $k = 0$ to represent the **point at infinity**, i.e. $\left(\frac{x}{0}, \frac{y}{0}\right)$ which is infinite.
- Generally, the $\mathbb{R}^n$ Euclidean space can be extended to a $\mathbb{P}^n$ projective space as homogeneous vectors.

Homogenous Coordinates



$$(x, y, 1) \equiv (2x, 2y, 2) \equiv (kx, ky, k)$$

We will see that this idea extends to lines and planes (in both 2D and 3D space)!

The 2D Projective Plane

- We will look at the **homogeneous notation** for lines $\mathbf{l}$ on a plane π , and the incidence relationship between points and lines.

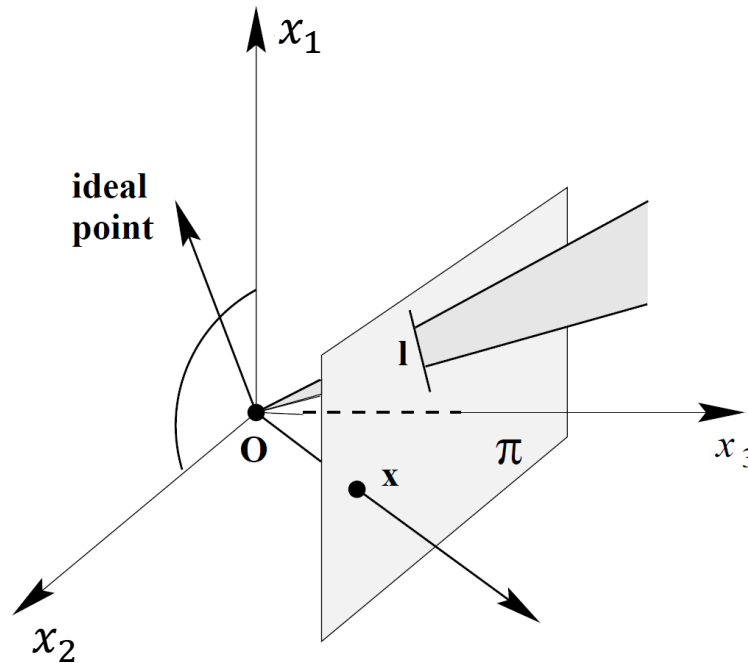


Image source: "Multiple View Geometry in Computer Vision", Richard Hartley and Andrew Zisserman

Homogeneous Representation of Lines and Points

- A **line in the plane** is represented by:

$$ax + by + c = 0$$

- Different choices of a , b and c giving rise to different lines.
- Thus, a line may naturally be represented by the vector $(a, b, c)^T$.

Homogeneous Representation of Lines and Points

- The correspondence between lines and vectors $(a, b, c)^T$ is **not one-to-one**.
- Since the lines $ax + by + c = 0$ and $(ka)x + (kb)y + (kc) = 0$ **are the same**, $\forall k \neq 0$.
- Thus $(a, b, c)^T$ and $k(a, b, c)^T$ represent **the same line**, for any non-zero k , i.e. equivalence class.
- **Note:** the vector $(0, 0, 0)^T$ **does not** correspond to any line.

Homogeneous Representation of Lines and Points

- A point $\mathbf{x} = (x, y)^T$ **lies on** the line $\mathbf{l} = (a, b, c)^T$ if and only if $ax + by + c = 0$, i.e.

$$(x, y, 1)(a, b, c)^T = (x, y, 1)\mathbf{l} = 0;$$

- Similarly, for any constant non-zero k ,

$$(kx, ky, k)(a, b, c)^T = k(x, y, 1)\mathbf{l} = (x, y, 1)\mathbf{l} = 0.$$

Homogeneous Representation of Lines and Points

- Hence, $(kx, ky, k)^T \in \mathbb{P}^2$ for varying values of k to be **a representation** of the point $(x, y)^T \in \mathbb{R}^2$, i.e.

$$\mathbf{x} = (x_1, x_2, x_3)^T \in \mathbb{P}^2 \equiv \left(\frac{x_1}{x_3}, \frac{x_2}{x_3} \right)^T \in \mathbb{R}^2.$$

Homogeneous Representation of Lines and Points

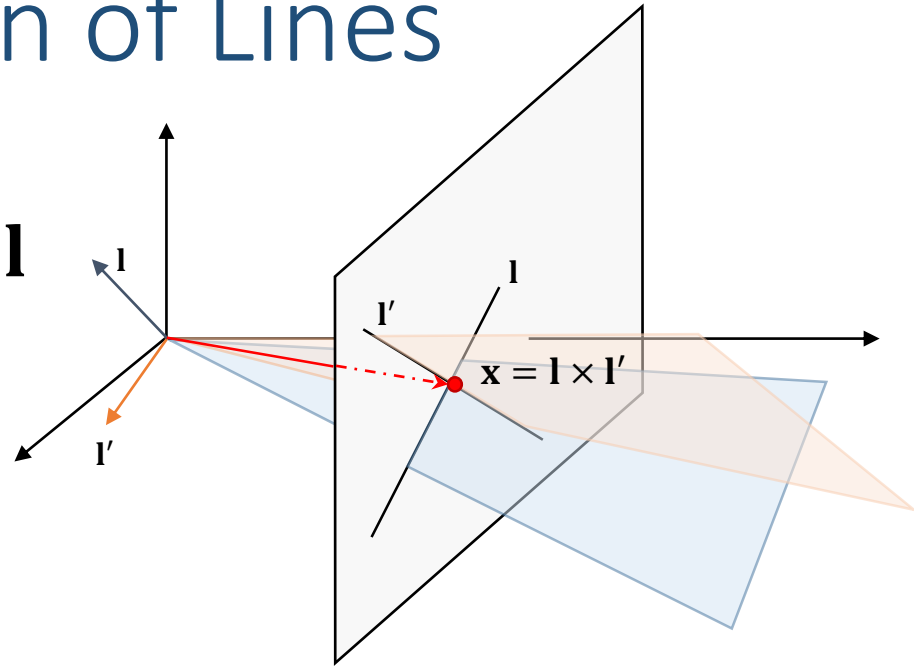
- More formally: The point $\mathbf{x}$ lies on the line $\mathbf{l}$ if and only if $\mathbf{x}^T \mathbf{l} = 0$.
- Note that the expression $\mathbf{x}^T \mathbf{l}$ is just the **inner or scalar product** of the two vectors $\mathbf{l}$ and $\mathbf{x}$; the scalar product:

$$\mathbf{x}^T \mathbf{l} = \mathbf{l}^T \mathbf{x} = \mathbf{x} \cdot \mathbf{l}$$

- **Degree of freedom (dof)**: a point has 2 dof – x and y coordinates; a line also has 2 dof – two independent ratios $\{a : b : c\}$.

Intersection of Lines

- The intersection of two lines $\mathbf{l}$ and $\mathbf{l}'$ is the point $\mathbf{x} = \mathbf{l} \times \mathbf{l}'$.



Proof:

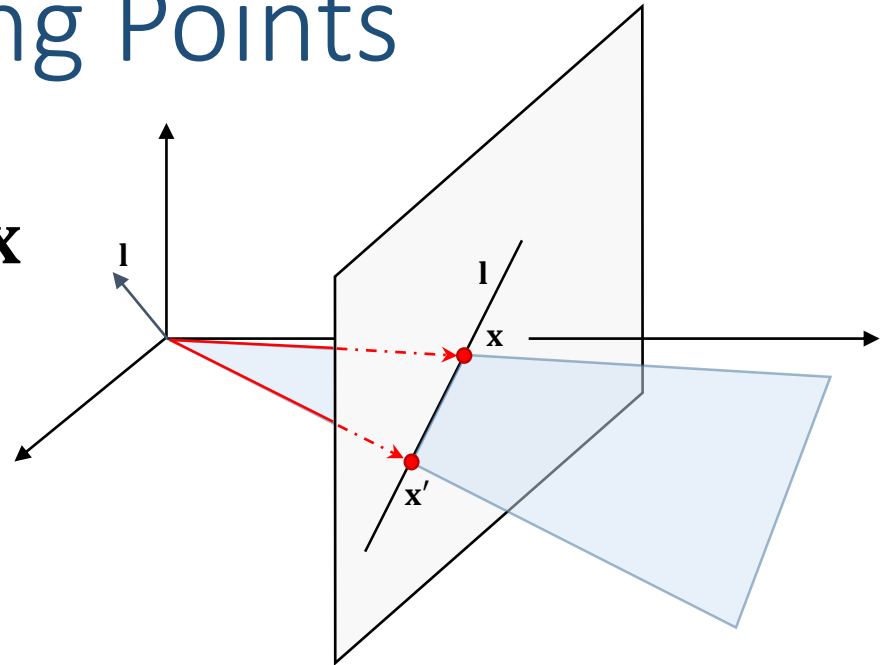
Given two lines $\mathbf{l} = (a, b, c)^T$ and $\mathbf{l}' = (a', b', c')^T$, the **triple scalar product identity** gives $\mathbf{l} \cdot (\mathbf{l} \times \mathbf{l}') = \mathbf{l}' \cdot (\mathbf{l} \times \mathbf{l}') = 0$, which we rewrite as:

$$\mathbf{l}^T \mathbf{x} = \mathbf{l}'^T \mathbf{x} = 0.$$

If $\mathbf{x}$ is thought of as **representing a point**, then $\mathbf{x}$ lies on both lines $\mathbf{l}$ and $\mathbf{l}'$, and hence is the intersection of the two lines. $\square$

Line Joining Points

- The line through two points $\mathbf{x}$ and $\mathbf{x}'$ is $\mathbf{l} = \mathbf{x} \times \mathbf{x}'$.



Proof:

Given two points $\mathbf{x}$ and $\mathbf{x}'$, the **triple scalar product identity** gives $\mathbf{x} \cdot (\mathbf{x} \times \mathbf{x}') = \mathbf{x}' \cdot (\mathbf{x} \times \mathbf{x}') = 0$, which we rewrite as:

$$\mathbf{x}^T \mathbf{l} = \mathbf{x}'^T \mathbf{l} = 0.$$

If $\mathbf{l}$ is thought of as **representing a line**, then $\mathbf{l}$ contains both points $\mathbf{x}$ and $\mathbf{x}'$, and hence is the line joining the two points.

Ideal Points and the Line at Infinity

Intersection of parallel lines

- Consider two parallel lines $ax + by + c = 0$ and $ax + by + c' = 0$, i.e. $\mathbf{l} = (a, b, c)^T$ and $\mathbf{l}' = (a, b, c')^T$.
- The intersection is $\mathbf{l} \times \mathbf{l}' = (c' - c)(b, -a, 0)^T$, i.e. $(b, -a, 0)^T$ ignoring the scale factor $(c' - c)$.
- $(b, -a, 0)^T$ is an **infinite point** and this implies that **parallel lines meet at infinity**.

Ideal Points and the Line at Infinity

Example:

Consider the two lines $x = 1$ and $x = 2$. Here the two lines are parallel, and consequently intersect “at infinity”.

In homogeneous notation the lines are $\mathbf{l} = (-1, 0, 1)^T$, $\mathbf{l}' = (-1, 0, 2)^T$, and their intersection point is:

$$\mathbf{x} = \mathbf{l} \times \mathbf{l}' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 0 & 1 \\ -1 & 0 & 2 \end{vmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

which is the point at infinity in the direction of the y -axis.

Ideal Points and the Line at Infinity

- The points $\mathbf{x} = (x_1, x_2, x_3)^T$ with last coordinate $x_3 = 0$ are known as **ideal points**, or **points at infinity**.
- The **set of all ideal points** may be written $(x_1, x_2, 0)^T$, with a particular point specified by the ratio $x_1 : x_2$.
- Note that this set lies on a single line, the **line at infinity**, denoted by the vector $\mathbf{l}_\infty = (0, 0, 1)^T$.

Proof:

$$(0, 0, 1)(x_1, x_2, 0)^T = 0.$$

Ideal Points and the Line at Infinity

- The parallel lines $\mathbf{l} = (a, b, c)^T$ and $\mathbf{l}' = (a, b, c')^T$ intersects $\mathbf{l}_\infty$ at the ideal point $(b, -a, 0)^T$ for all c 's.
- In inhomogeneous notation $(b, -a)^T$ is a vector **tangent** to the line, and **orthogonal** to the line normal (a, b) , and so represents the line's *direction*.
- As the line's direction varies, the ideal point $(b, -a, 0)^T$ varies over $\mathbf{l}_\infty$.
- Hence, the line at infinity can be thought of as **the set of directions of lines** in the plane.

Duality principle

- Notice how the role of points and lines **may be interchanged** in:
 1. Incidence equations, i.e. $\mathbf{l}^T \mathbf{x} = 0$ and $\mathbf{x}^T \mathbf{l} = 0$.
 2. Intersection of two lines and the line through two points, i.e. $\mathbf{x} = \mathbf{l} \times \mathbf{l}'$ and $\mathbf{l} = \mathbf{x} \times \mathbf{x}'$.
- These observations lead to the **duality principle**.

Duality principle

- **Duality principle.** To any theorem of 2-dimensional projective geometry there corresponds a dual theorem, which may be derived **by interchanging the roles of points and lines** in the original theorem.
- Consequently, it **not necessary** to prove the dual of a given theorem once the original theorem has been proven.
- The proof of the dual theorem **will be** the dual of the proof of the original theorem.

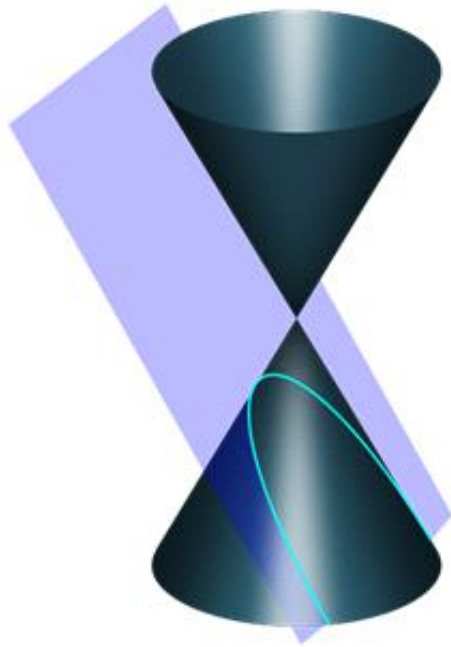
Conics and Dual Conics

- A conic is a curve described by a **second-degree equation** in the plane.
- In Euclidean geometry, conics are of three main types: **hyperbola, ellipse, and parabola**.
- These three types of conic arise as conic sections generated by **planes of differing orientation**.
- **Note:** there are also **degenerate conics**, which we will define later.

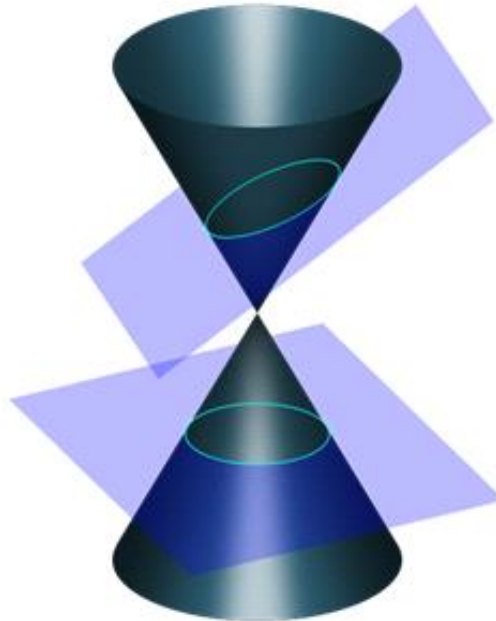
Conics and Dual Conics

Types of conics:

Parabola



Ellipse and Circle



Hyperbola



Image source: https://en.wikipedia.org/wiki/Conic_section

Conics and Dual Conics

- The **equation of a conic** in inhomogeneous coordinates is a polynomial of degree 2, i.e.

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

- “**Homogenizing**” this by the replacements: $x \rightarrow \frac{x_1}{x_3}$, $y \rightarrow \frac{x_2}{x_3}$ gives

$$ax_1^2 + bx_1x_2 + cx_2^2 + dx_1x_3 + ex_2x_3 + fx_3^2 = 0$$

Conics and Dual Conics

- Or in **matrix form**: $\mathbf{x}^T \mathbf{C} \mathbf{x} = 0$, where $\mathbf{C}$ is symmetric and given by:

$$\mathbf{C} = \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}.$$

- $\mathbf{C}$ is a **homogeneous representation** of a conic.
- Only the ratios of the matrix elements are important, multiplying $\mathbf{C}$ by a non-zero scalar has no effect.

Conics and Dual Conics

$$\mathbf{x}^T \mathbf{C} \mathbf{x} = 0, \quad \mathbf{C} = \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}.$$

- The conic has **five degrees of freedom** which can be thought of as the ratios $\{a : b : c : d : e : f\}$.
- Or equivalently the **six elements** of a symmetric matrix **less one for scale**.

Five Points Define a Conic

- Each point $\mathbf{x}_i = (x_i, y_i)$ places one constraint on the conic coefficients:

$$ax_i^2 + bx_iy_i + cy_i^2 + dx_i + ey_i + f = 0.$$

- This constraint can be written as:

$$\begin{pmatrix} x_i^2 & x_iy_i & y_i^2 & x_i & y_i & 1 \end{pmatrix} \mathbf{c} = 0$$

- where $\mathbf{c} = (a, b, c, d, e, f)^T$ is the conic C represented as a 6-vector.

Five Points Define a Conic

- Stacking the constraints from **five points** we obtain

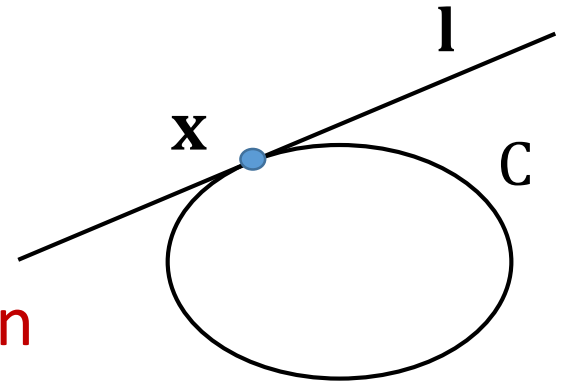
$$\begin{bmatrix} x_1^2 & x_1y_1 & y_1^2 & x_1 & y_1 & 1 \\ x_2^2 & x_2y_2 & y_2^2 & x_2 & y_2 & 1 \\ x_3^2 & x_3y_3 & y_3^2 & x_3 & y_3 & 1 \\ x_4^2 & x_4y_4 & y_4^2 & x_4 & y_4 & 1 \\ x_5^2 & x_5y_5 & y_5^2 & x_5 & y_5 & 1 \end{bmatrix} \mathbf{c} = \mathbf{0}$$

- The conic is the **null vector** of this 5×6 matrix.
- This shows that a conic is determined uniquely (up to scale) by five points in general position.

Conics and Dual Conics

Tangent lines to conics:

The line $\mathbf{l}$ tangent to C at a point $\mathbf{x}$ on C is given by $\mathbf{l} = C\mathbf{x}$.



Proof:

The line $\mathbf{l} = C\mathbf{x}$ passes through $\mathbf{x}$, since $\mathbf{l}^T \mathbf{x} = \mathbf{x}^T C\mathbf{x} = 0$. If $\mathbf{l}$ has **one-point contact** with the conic, then it is a tangent, and we are done.

□

Conics and Dual Conics

- The conic C defined as far is more properly termed a **point conic**, as it defines an equation on points.
- There is also a **dual (line) conic** which defines an equation on lines denoted as C^* (3x3 matrix).
- A line $\mathbf{l}$ *tangent* to the conic C satisfies $\mathbf{l}^T C^* \mathbf{l} = 0$.
- A dual conic has **five degrees of freedom** and can be computed from five lines.

Conics and Dual Conics

- For a **non-singular** symmetric matrix $C^* = C^{-1}$ (up to scale).

Proof:

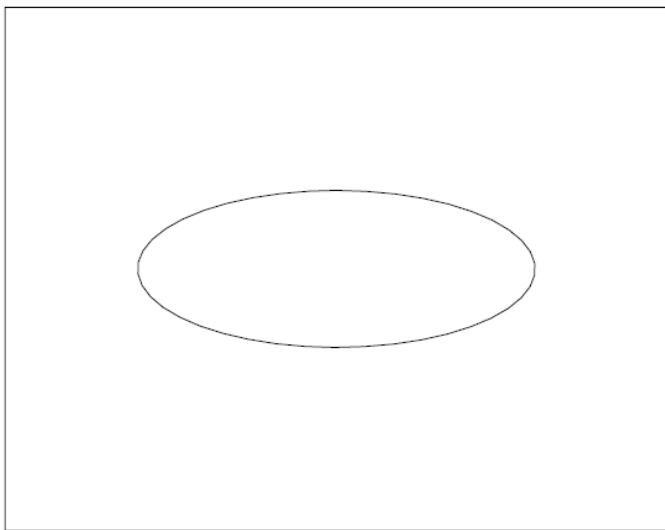
A point $\mathbf{x}$ on C , the tangent is $\mathbf{l} = C\mathbf{x}$ and this implies $\mathbf{x} = C^{-1}\mathbf{l}$, i.e. $C^* = C^{-1}$ and $\mathbf{x} = C^*\mathbf{l}$.

Furthermore, since $\mathbf{x}$ satisfies $\mathbf{x}^T C \mathbf{x} = 0$, we obtain $(C^{-1}\mathbf{l})^T C (C^{-1}\mathbf{l}) = \mathbf{l}^T C^{-1} \mathbf{l} = 0$, where $C^{-T} = C^{-1}$; we can write as $\mathbf{l}^T C^* \mathbf{l} = 0$.

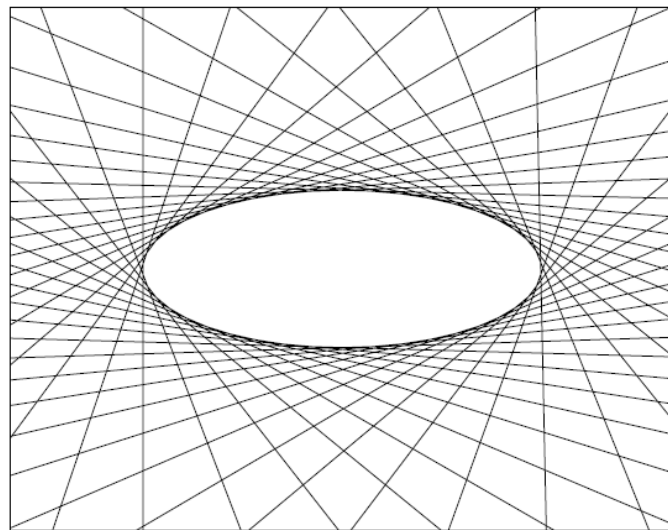
□

Conics and Dual Conics

- Dual conics are also known as **conic envelopes**:



Points $\mathbf{x}$ satisfying $\mathbf{x}^T \mathbf{C} \mathbf{x} = 0$ lie on a point conic.



Lines $\mathbf{l}$ satisfying $\mathbf{l}^T \mathbf{C}^* \mathbf{l} = 0$ are tangent to the point conic $\mathbf{C}$. The conic $\mathbf{C}$ is the envelope of the lines $\mathbf{l}$.

Image source: “Multiple View Geometry in Computer Vision”, Richard Hartley and Andrew Zisserman

Degenerate Conics

- Suppose that $\mathbf{l}$ meets the conic in **another point** $\mathbf{y}$, then $\mathbf{y}^T \mathbf{C} \mathbf{y} = 0$ and $\mathbf{x}^T \mathbf{C} \mathbf{y} = \mathbf{l}^T \mathbf{y} = 0$.
- From this it follows that $(\mathbf{x} + \alpha \mathbf{y})^T \mathbf{C} (\mathbf{x} + \alpha \mathbf{y}) = 0$ for all α .
- This means that the **whole line** $\mathbf{l} = \mathbf{C} \mathbf{x}$ joining $\mathbf{x}$ and $\mathbf{y}$ **lies on the conic** $\mathbf{C}$, which is therefore degenerate.

Degenerate Conics

$$(\mathbf{x} + \alpha \mathbf{y})^T C (\mathbf{x} + \alpha \mathbf{y}) = 0$$

$$\mathbf{l} = C\mathbf{x} = C\mathbf{y} \text{ and } \mathbf{l}^T \mathbf{y} = \mathbf{l}^T \mathbf{x} = 0; \text{ rank}(C) < 3$$

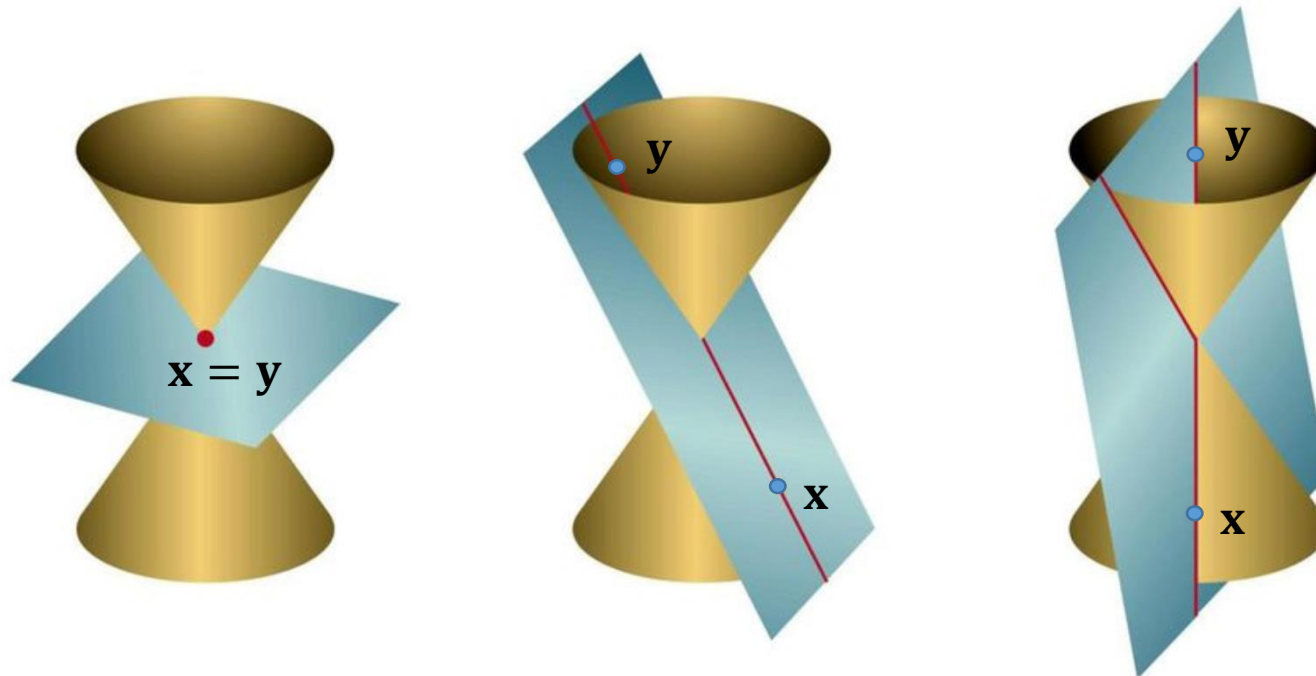


Image source: <https://slideplayer.com/slide/12844330/>

Degenerate Conics

- If the matrix C is **not of full rank**, then the conic is termed degenerate.
- Degenerate conics include **two lines (rank 2)**, and a **repeated line (rank 1)**.

Example: The conic $C = \mathbf{l} \mathbf{m}^T + \mathbf{m} \mathbf{l}^T$ is composed of two lines $\mathbf{l}$ and $\mathbf{m}$. Points on $\mathbf{l}$ satisfy $\mathbf{l}^T \mathbf{x} = 0$, and are on the conic since

$$\mathbf{x}^T C \mathbf{x} = (\mathbf{x}^T \mathbf{l})(\mathbf{m}^T \mathbf{x}) + (\mathbf{x}^T \mathbf{m})(\mathbf{l}^T \mathbf{x}) = 0.$$

We can see geometrically that this is **two straight lines**:

$$\begin{aligned}\mathbf{x}^T \mathbf{l} &= \mathbf{l}^T \mathbf{x} = l_1 x_1 + l_2 x_2 + l_3 \\ \mathbf{m}^T \mathbf{x} &= \mathbf{x}^T \mathbf{m} = m_1 x_1 + m_2 x_2 + m_3\end{aligned}$$

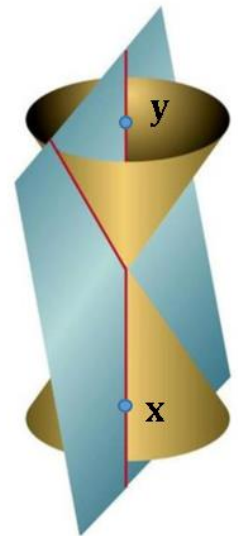


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Degenerate Conics

- If the matrix C is **not of full rank**, then the conic is termed degenerate.
- Degenerate conics include **two lines (rank 2)**, and a **repeated line (rank 1)**.

Example: Similar method can be used to show that $C = \mathbf{l}\mathbf{l}^T + \mathbf{l}\mathbf{l}^T$ contains a repeated line. We have

$$\mathbf{x}^T C \mathbf{x} = (\mathbf{x}^T \mathbf{l})(\mathbf{l}^T \mathbf{x}) + (\mathbf{x}^T \mathbf{l})(\mathbf{l}^T \mathbf{x}) = 0,$$

which consists of

$$\mathbf{x}^T \mathbf{l} = \mathbf{l}^T \mathbf{x} = l_1 x_1 + l_2 x_2 + l_3.$$

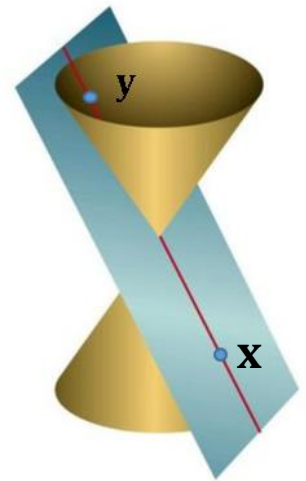


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Degenerate Dual Conics

- Degenerate **dual (line) conics** include **two points (rank 2)**, and a **repeated point (rank 1)**.

Example:

The line conic $C^* = \mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T$ has rank 2 and consists of lines passing through either of the two points $\mathbf{x}$ and $\mathbf{y}$.

Similar formulation can be done for **rank 1** line conic of **repeated points**.

Note that for matrices that are not invertible $(C^*)^* \neq C$.

Planar Projective Transformations

- 2D projective geometry is the study of properties of the projective plane $\mathbb{P}^2$ that are **invariant under a group of transformations** known as projectivities.
- A **projectivity** is an **invertible mapping** h from $\mathbb{P}^2$ to itself such that three points $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ **lie on the same line** if and only if $h(\mathbf{x}_1)$, $h(\mathbf{x}_2)$ and $h(\mathbf{x}_3)$ do.
- A projectivity is also called a **collineation**, a **projective transformation** or a **homography**.

Planar Projective Transformations

Theorem:

A mapping $h : \mathbb{P}^2 \rightarrow \mathbb{P}^2$ is a projectivity if and only if there exists a **non-singular** 3×3 matrix H such that for any point in $\mathbb{P}^2$ represented by a vector $\mathbf{x}$ it is true that **$h(\mathbf{x}) = H\mathbf{x}$** .

Partial Proof:

Let $\mathbf{x}_1, \mathbf{x}_2$ and $\mathbf{x}_3$ **lie on a line** $\mathbf{l}$. Thus $\mathbf{l}^T \mathbf{x}_i = 0$ for $i = 1, \dots, 3$. Let H be a **non-singular** 3×3 matrix.

We can verify that $\mathbf{l}^T H^{-1} H \mathbf{x}_i = 0$. Thus, the points $H \mathbf{x}_i$ all lie on the line $H^{-T} \mathbf{l}$, and hence **collinearity is preserved** by the transformation.

Note: We skip the converse which is harder to prove, i.e. each projectivity arises in this way.

Planar Projective Transformations

- We now define planar projective transformation as:

$$\begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad \text{or} \quad \mathbf{x}' = H\mathbf{x}.$$

- Properties of H :

1. **Non-singular** 3×3 matrix;
2. **Homogeneous** matrix since only the ratio of the matrix elements is significant;
3. **Eight degrees of freedom**, i.e. eight independent ratios amongst the nine elements of H .

Planar Projective Transformations

- **Central projection** maps points on one plane to points on another plane.
- And represented by a **linear mapping** of homogeneous coordinates $\mathbf{x}' = H\mathbf{x}$.

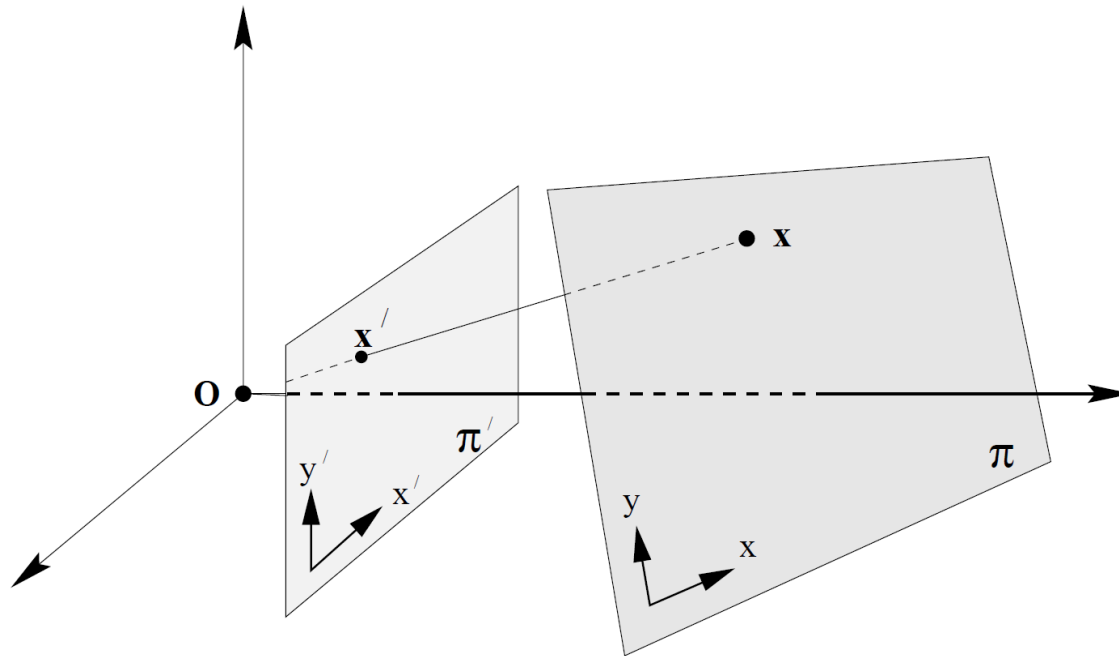


Image source: "Multiple View Geometry in Computer Vision", Richard Hartley and Andrew Zisserman

Planar Projective Transformations

- Examples of a projective transformation $x' = Hx$, arising in **perspective images**.

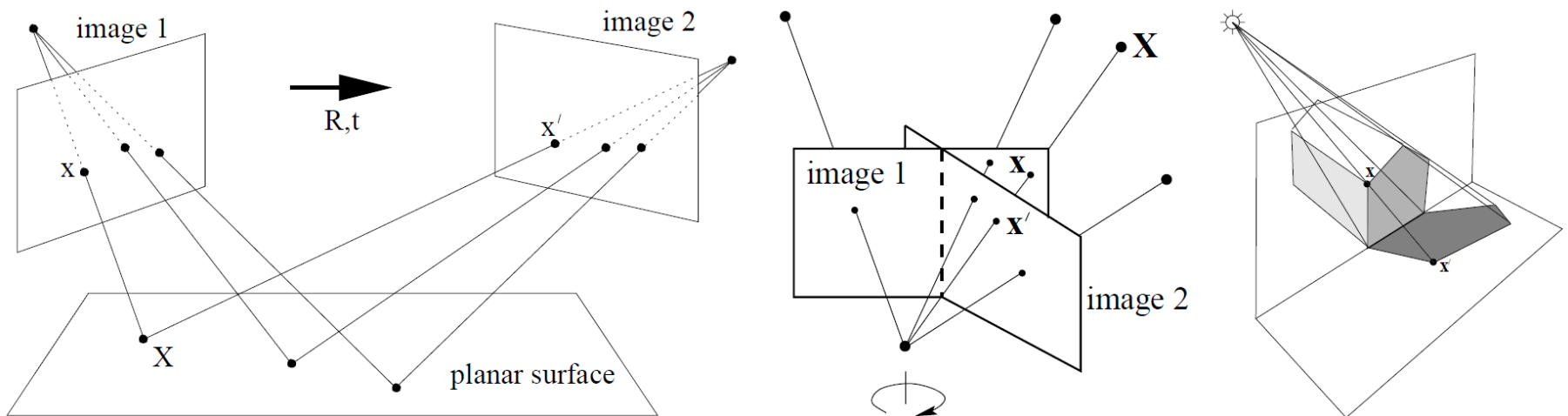


Image source: "Multiple View Geometry in Computer Vision", Richard Hartley and Andrew Zisserman

Transformations of Lines and Conics

- We have seen earlier that:

If points $\mathbf{x}_i$ lie on a line $\mathbf{l}$, then the transformed points $\mathbf{x}'_i = H\mathbf{x}_i$ under a projective transformation lie on the line $\mathbf{l}' = H^{-T}\mathbf{l}$.

In this way, **incidence of points on lines is preserved**, since $\mathbf{l}'^T \mathbf{x}'_i = \mathbf{l}^T H^{-1} H \mathbf{x}_i = 0$.

- This means that under the point transformation $\mathbf{x}' = H\mathbf{x}$, **a line transforms as:**

$$\mathbf{l}' = H^{-T}\mathbf{l}, \text{ or } \mathbf{l}'^T = \mathbf{l}^T H^{-1}.$$

Transformations of Lines and Conics

- Under a point transformation $\mathbf{x}' = H\mathbf{x}$, a conic C transforms to $C' = H^{-T}CH^{-1}$.
- Under a point transformation $\mathbf{x}' = H\mathbf{x}$, a dual conic C^* transforms to $C^{*'} = HC^*H^T$.

Proof:

Under a point transformation $\mathbf{x}' = H\mathbf{x}$,

$$\begin{aligned}\mathbf{x}^T C \mathbf{x} &= \mathbf{x}'^T [H^{-1}]^T C H^{-1} \mathbf{x}' \\ &= \mathbf{x}'^T H^{-T} C H^{-1} \mathbf{x}'\end{aligned}$$

which is a quadratic form $\mathbf{x}'^T C' \mathbf{x}'$ with $C' = H^{-T}CH^{-1}$.

Hierarchy of Transformations: Isometries

- Isometries are transformations of the plane $\mathbb{R}^2$ that **preserve Euclidean distance**, and represented as

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} \epsilon \cos \theta & -\sin \theta & t_x \\ \epsilon \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}, \text{ or}$$

$$\mathbf{x}' = \mathbf{H}_E \mathbf{x} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \mathbf{x}$$

- where $\epsilon = \pm 1$.

Hierarchy of Transformations: Isometries

- If $\epsilon = 1$, then the isometry is **orientation-preserving** and is a **Euclidean transformation** (rotation matrix R and translation t).
- If $\epsilon = -1$, then the isometry **reverses orientation**, e.g. reflection.
- **Invariants:** Length, angle and area.

Hierarchy of Transformations: Similarity

- Similarity transformation is an isometry composed with an **isotropic scaling**, and represented as

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} s \cos \theta & -s \sin \theta & t_x \\ s \sin \theta & s \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}, \text{ or}$$

$$\mathbf{x}' = \mathbf{H}_s \mathbf{x} = \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \mathbf{x}$$

- where the scalar s represents the **isotropic scaling**.

Hierarchy of Transformations: Similarity

- A similarity transformation is also known as an *equi-form transformation*, because it preserves “shape” (form).
- H_s has *four degrees of freedom* (3 isometry + 1 scale) and can be computed from two point correspondences.
- **Invariants:** Angles, ratio of two lengths and ratio of areas.

Hierarchy of Transformations: Affinity

- Affine transformation is a **non-singular linear transformation** followed by a **translation**, and represented as

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}, \text{ or } \mathbf{x}' = H_A \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \mathbf{x}$$

- where $\mathbf{A}$ is a 2×2 **non-singular** matrix.
- H_A has **six degrees of freedom** and can be computed from three point correspondences.
- Invariants:** parallel lines, ratio of lengths of parallel line segments and ratio of areas.

Hierarchy of Transformations: Affinity

- The **affine matrix A** can always be decomposed as:

$$A = R(\theta) R(-\phi) D R(\phi)$$

- $R(\theta)$ and $R(\phi)$ are **rotations** by θ and ϕ respectively, and D is a diagonal matrix:

$$D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}.$$

- This decomposition follows directly from the **Singular Value Decomposition (SVD)**:

$$A = UDV^T = (UV^T)(VDV^T) = R(\theta)(R(-\phi)DR(\phi)).$$

- Since U and V are **orthogonal matrices**.

Hierarchy of Transformations: Affinity

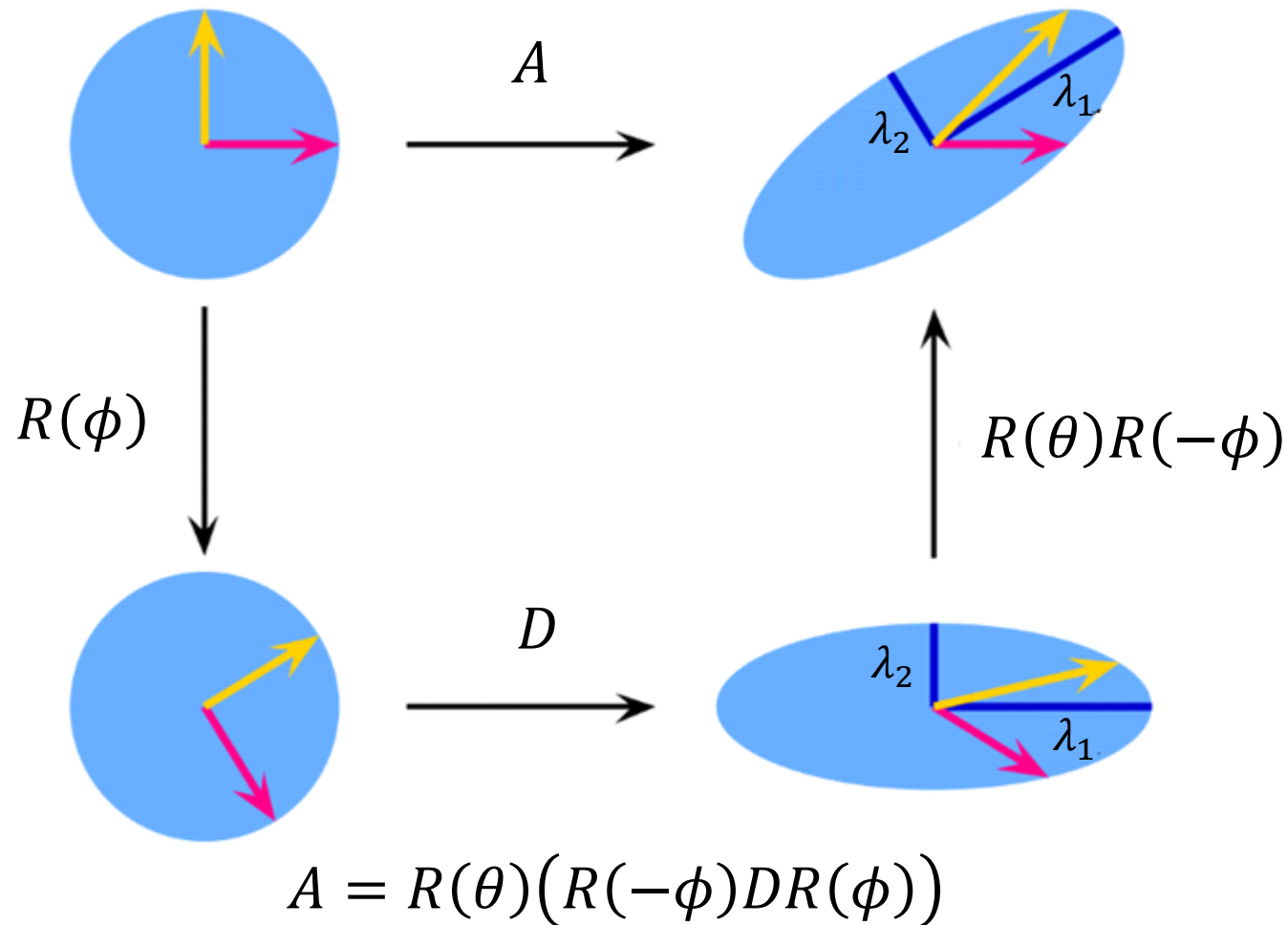


Image modified from: https://en.wikipedia.org/wiki/Singular_value_decomposition

Hierarchy of Transformations: Projective

- Projective transformation is a general **non-singular linear transformation** of homogeneous coordinates, and represented as

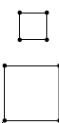
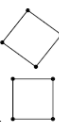
$$\mathbf{x}' = H_P \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^T & v \end{bmatrix} \mathbf{x}$$

- where the vector $\mathbf{v} = (v_1, v_2)^T$ and v can be 0.
- H_p has nine elements with only their ratio significant, so the transformation has **eight degrees of freedom**.

Hierarchy of Transformations: Projective

- Note, it is **not always possible** to scale the matrix such that v is unity since v might be zero.
- A projective transformation between two planes can be computed from **four point correspondences**, with no three collinear on either plane.
- **Not possible to distinguish** between orientation preserving and orientation reversing projectivities in $\mathbb{P}^2$.
- **Invariants:** order of contact, tangency (2 pt contact) and cross ratio (details later).

Hierarchy of Transformations

Group	Matrix	Distortion	Invariant properties
Projective 8 dof	$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$		Concurrency, collinearity, order of contact : intersection (1 pt contact); tangency (2 pt contact); inflections (3 pt contact with line); tangent discontinuities and cusps. cross ratio (ratio of ratio of lengths).
Affine 6 dof	$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Parallelism, ratio of areas, ratio of lengths on collinear or parallel lines (e.g. midpoints), linear combinations of vectors (e.g. centroids). The line at infinity, l_∞ (more later).
Similarity 4 dof	$\begin{bmatrix} sr_{11} & sr_{12} & t_x \\ sr_{21} & sr_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Ratio of lengths, angle. The circular points, I, J (more later).
Euclidean 3 dof	$\begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Length, area

Source: Page 44, Table 2.1, "Multiple View Geometry in Computer Vision", Richard Hartley and Andrew Zisserman

Decomposition of a Projective Transformation

- A projective transformation can be decomposed into a **chain of transformations**:

$$H = H_S H_A H_P = \begin{bmatrix} sR & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} K & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} I & \mathbf{0} \\ \mathbf{v}^T & v \end{bmatrix} = \begin{bmatrix} A & \mathbf{t} \\ \mathbf{v}^T & v \end{bmatrix}$$

- A **non-singular matrix** given by $A = sRK + \mathbf{t}\mathbf{v}^T$.
- K an **upper-triangular matrix** normalized as $\det(K) = 1$.
- Decomposition is valid provided $v \neq 0$, and is unique if s is chosen positive.
- We will see that this decomposition **preserves geometric properties** of $\mathbf{l}_\infty$ and the circular points (Lecture 3).

Projective Geometry of 1D (Line), $\mathbb{P}^1$

- We denote a **point on the line** as the homogeneous coordinates $\bar{\mathbf{x}}' = (x_1, x_2)^T$.
- $x_2 = 0$ is an **ideal point** of the line.
- A projective transformation of a line is represented by a 2×2 homogeneous matrix,

$$\bar{\mathbf{x}}' = H_{2 \times 2} \bar{\mathbf{x}}$$

- $H_{2 \times 2}$ has **3 dof** corresponding to 4 elements less one for over scaling, and can be computed from 3 points.

Projective Geometry of 1D (Line), $\mathbb{P}^1$

The Cross Ratio

- The cross ratio is the basic **projective invariant** of $\mathbb{P}^1$. Given 4 points $\bar{\mathbf{x}}_i$ the *cross ratio* is defined as:

$$\text{Cross}(\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2, \bar{\mathbf{x}}_3, \bar{\mathbf{x}}_4) = \frac{|\bar{\mathbf{x}}_1 \bar{\mathbf{x}}_2| |\bar{\mathbf{x}}_3 \bar{\mathbf{x}}_4|}{|\bar{\mathbf{x}}_1 \bar{\mathbf{x}}_3| |\bar{\mathbf{x}}_2 \bar{\mathbf{x}}_4|},$$

where

$$|\bar{\mathbf{x}}_i \bar{\mathbf{x}}_j| = \det \begin{bmatrix} x_{i1} & x_{j1} \\ x_{i2} & x_{j2} \end{bmatrix}.$$

- If each point $\bar{\mathbf{x}}_i$ is a finite point and $x_2 = 1$, then $|\bar{\mathbf{x}}_i \bar{\mathbf{x}}_j|$ represents the **signed distance** from $\bar{\mathbf{x}}_i$ to $\bar{\mathbf{x}}_j$.
- Definition of the cross ratio is also valid if one of the points $\bar{\mathbf{x}}_i$ is an **ideal point**.

Projective Geometry of 1D (Line), $\mathbb{P}^1$

- The value of the cross ratio is **invariant under any projective transformation** of the line: if $\bar{x}' = H_{2 \times 2} \bar{x}$ then

$$\text{Cross}(\bar{x}'_1, \bar{x}'_2, \bar{x}'_3, \bar{x}'_4) = \text{Cross}(\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4).$$

Exercise: Prove it!

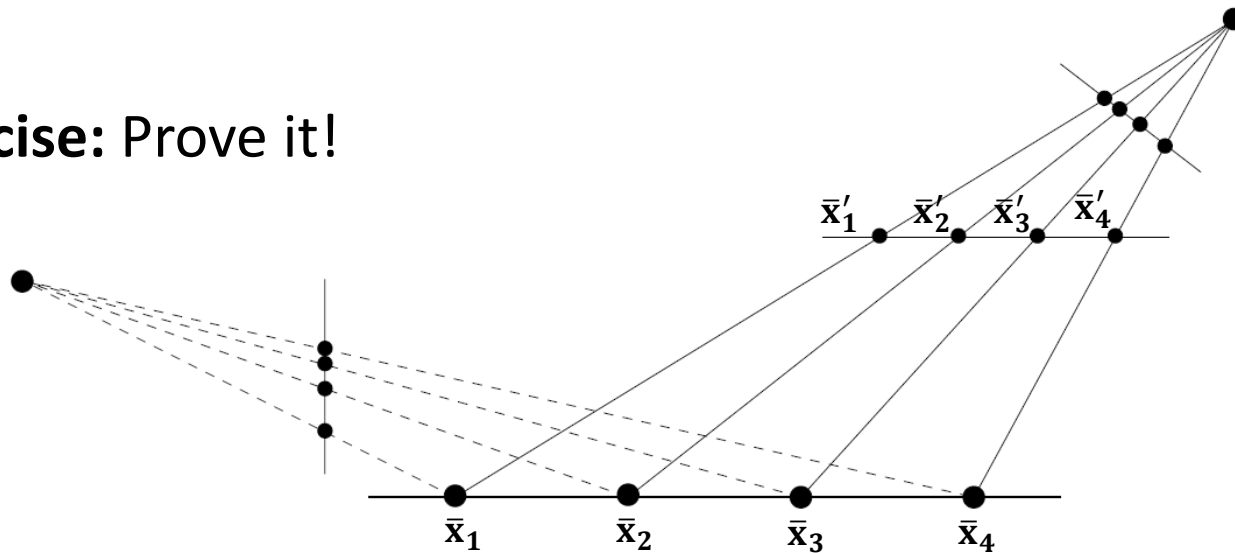


Image source: "Multiple View Geometry in Computer Vision", Richard Hartley and Andrew Zisserman

Projective Geometry of 1D (Line), $\mathbb{P}^1$

Concurrent Lines

- A configuration of concurrent lines is **dual to** collinear points on a line, i.e. concurrent lines on a plane are also in $\mathbb{P}^1$.
- Four concurrent lines l_i intersect the line l in the four points $\bar{x}_i$.
- The cross ratio of these lines is an **invariant to** projective transformations of the plane.
- Its value is given by the **cross ratio of the points**, $\text{Cross}(\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4)$.

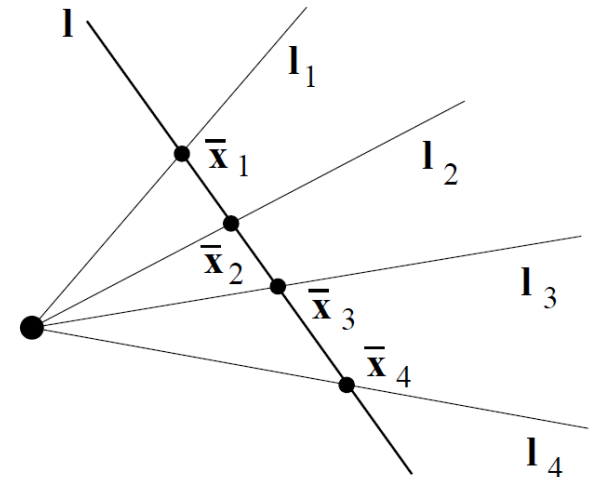


Image source: "Multiple View Geometry in Computer Vision", Richard Hartley and Andrew Zisserman

Projective Geometry of 1D (Line), $\mathbb{P}^1$

Concurrent Lines

- Coplanar points $\mathbf{x}_i$ are imaged onto a line $\mathbf{l}$ (also in the plane) by a projection with centre $\mathbf{C}$.
- May be thought of as representing projection of points in $\mathbb{P}^2$ into a 1-dimensional image.
- In particular, the line $\mathbf{l}$ represents an 1D analogue of the image plane.
- The cross ratio of the image points $\bar{\mathbf{x}}_i$ is invariant to the position of the image line $\mathbf{l}$.

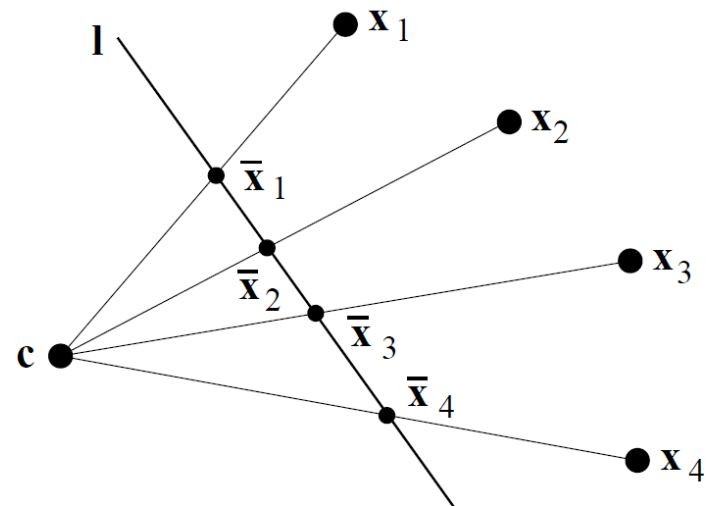


Image source: "Multiple View Geometry in Computer Vision", Richard Hartley and Andrew Zisserman

Summary

- We have looked at how to:
 1. Explain the difference between **Euclidean and Projective geometry**.
 2. Use **homogenous coordinates** to represent **points, lines and conics** in the projective space.
 3. Describe the **duality relation** between lines and points, and conics and dual conics on a plane.
 4. Apply the **hierarchy of transformations** on points, lines and conics.